

How Group Theory Secures Online Banking and E-Commerce

A Study of RSA Cryptography

Presented by:

Geetika Pachauri
Anusha Anindita
Harshil Pansala
Shirsha Pattanaik
Pearl Mendapara

Instructor: Professor Pradeep Kumar

Department: Mathematics, Mahindra University

Date: 25 November 2025

From Abstract to Applied

Mathematical elegance creates practical security.



Abstract Algebra Emerges (1800s)

Laying the theoretical groundwork for structure and symmetry in mathematics.



Group Theory Formalized

Defining operations and properties crucial for cryptographic applications.



Number Theory & Groups Converge

The interplay of discrete numbers and group structures becomes apparent.



RSA Algorithm Invented (1977)

Revolutionizing secure communication with public-key cryptography.



Digital Security Revolution

Transforming how we protect information and transactions online.



Your Data Protected Today

Securing online banking, e-commerce, and personal privacy across the globe.

Today's Journey



Introduction to RSA &
Cryptography



Group Theory Fundamentals



Mathematical Foundations of
RSA



RSA Algorithm Walkthrough



Worked Example



Security Analysis



Real-World Applications



Future



Conclusion

Cryptography: The Art of Secure Communication

Cryptography is the science of protecting information through advanced mathematical techniques, ensuring that data remains confidential and unaltered during transmission and storage. It transforms sensitive data into an unreadable format, only accessible to authorized parties.

Confidentiality

Keeps information private from unauthorized access.

Integrity

Ensures data has not been altered or tampered with.

Authentication

Verifies the identity of sender and receiver.

Non-repudiation

Prevents denying participation in a transaction.

From ancient ciphers to modern digital security, cryptography has evolved to safeguard our digital lives.

Ancient Ciphers

Simple substitution and transposition techniques.

Early Digital

DES, Lucifer, and the dawn of computer-based encryption.

Mechanical Enigma

Complex rotor machines during World Wars.

Modern Cryptography

Public-key systems like RSA, elliptic curves, and quantum-resistant algorithms.

RSA: The Backbone of Digital Security

The RSA algorithm, named after its inventors Rivest, Shamir, and Adleman, emerged in 1977 as a foundational public-key cryptosystem. Its core principle leverages mathematical operations that are easy to perform in one direction but computationally infeasible to reverse without specific knowledge.



Public Key

A publicly shared key is used by anyone to encrypt messages.



Private Key

A secret key, known only to the recipient, is used for decryption.



Factorization

Security is based on the extreme difficulty of factoring very large numbers.

Part I: Mathematical Foundations

Understanding Groups

Delving into the elegant world of group theory, the bedrock of RSA's security.

Group Theory: The Building Blocks

□ A group $(G, *)$ is a set G with a binary operation $*$ that satisfies four fundamental axioms.

The Four Axioms:

1. **Closure:** For all $a, b \in G$, $a * b \in G$.
2. **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
3. **Identity Element:** There exists an element $e \in G$ such that $a * e = a$ for all $a \in G$.
4. **Inverse Element:** For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = e$.

Groups Around Us

Group theory describes fundamental symmetries and structures found in everyday life and nature. Let's explore two common examples.

Clock Arithmetic (Modular Arithmetic)

The hours on a clock (0-11) form a group under addition modulo 12. **0** is the identity element, and every hour has an inverse (e.g., $9 + 3 = 0 \pmod{12}$). If it's 9 o'clock and 5 hours pass, it's 2 o'clock ($9 + 5 = 14 \equiv 2 \pmod{12}$).

Rotations of a Square

The four rotations of a square (0° , 90° , 180° , 270°) form a group under composition. The **0° rotation** is the identity, and each rotation has an inverse (e.g., 90° and 270° are inverses). For example, a 90° rotation followed by another 90° rotation is equivalent to a 180° rotation.

These examples illustrate how group theory's axioms apply to practical systems, providing a framework for understanding symmetry and structure.

The Group RSA Lives In

The Multiplicative Group Z_n^*

The security of RSA cryptography fundamentally relies on the properties of a specific mathematical structure: the multiplicative group of integers modulo n , denoted as Z_n^* .

Definition of Z_n^*

The set Z_n^* comprises all positive integers a such that $1 \leq a < n$ and the greatest common divisor of a and n is 1, i.e., $\gcd(a, n) = 1$.

Practical Meaning

This means Z_n^* includes all numbers less than n that are **coprime** to n (they share no common prime factors with n).

Example: Z_{10}^*

For $n = 10$, the integers less than 10 that are coprime to 10 are $\{1, 3, 7, 9\}$. This set forms a group under multiplication modulo 10.

Group Properties in Z_{10}^*

Verifying the Axioms

Let's confirm how the multiplicative group Z_{10}^* , consisting of $\{1, 3, 7, 9\}$ under multiplication modulo 10, satisfies the four fundamental group axioms. Understanding these properties is crucial for appreciating RSA's security.



Identity Element

The element 1 is the identity; multiplying any group member by 1 returns the same number (e.g., $3 \times 1 \equiv 3 \pmod{10}$).



Closure

Multiplying any two numbers in Z_{10}^* results in another number within the same group (e.g., $3 \times 7 \equiv 1 \pmod{10}$).



Inverse Element

Every number has a unique inverse in the group; when multiplied, they yield the identity element 1 (e.g., $3 \times 7 \equiv 1 \pmod{10}$).



Associativity

The order of operations for multiplication modulo 10 does not change the outcome (e.g., $(3 \times 7) \times 9 \equiv 3 \times (7 \times 9) \pmod{10}$).

 **Key Insight:** These inherent group properties ensure that cryptographic operations within RSA, such as encryption and decryption, are predictable and reversible.

Counting Elements: $\phi(n)$

Euler's Totient Function, $\phi(n)$, counts integers less than n that are coprime to it. This function determines the size of the multiplicative group Z_n^* , a core element in RSA's security.

Definition

$\phi(n)$ = the number of positive integers $a < n$ such that $\gcd(a, n) = 1$.

Key Formula

For distinct primes p and q , as used in RSA, $\phi(pq) = (p - 1)(q - 1)$.

Example: $\phi(10)$

Given $n = 10$ (2×5):

$$\phi(10) = (2 - 1)(5 - 1) = 1 \times 4 = 4.$$

This confirms $Z_{10}^* = \{1, 3, 7, 9\}$ indeed has 4 elements.

The Crown Jewel of Group Theory: Euler's Theorem

Euler's Theorem generalizes Fermat's Little Theorem and forms the mathematical basis for RSA encryption and decryption. It governs exponentiation in modular arithmetic.

Theorem

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

For any integer a coprime to n (i.e., $a \in Z_n^*$).

📌 This is WHY RSA decryption works!

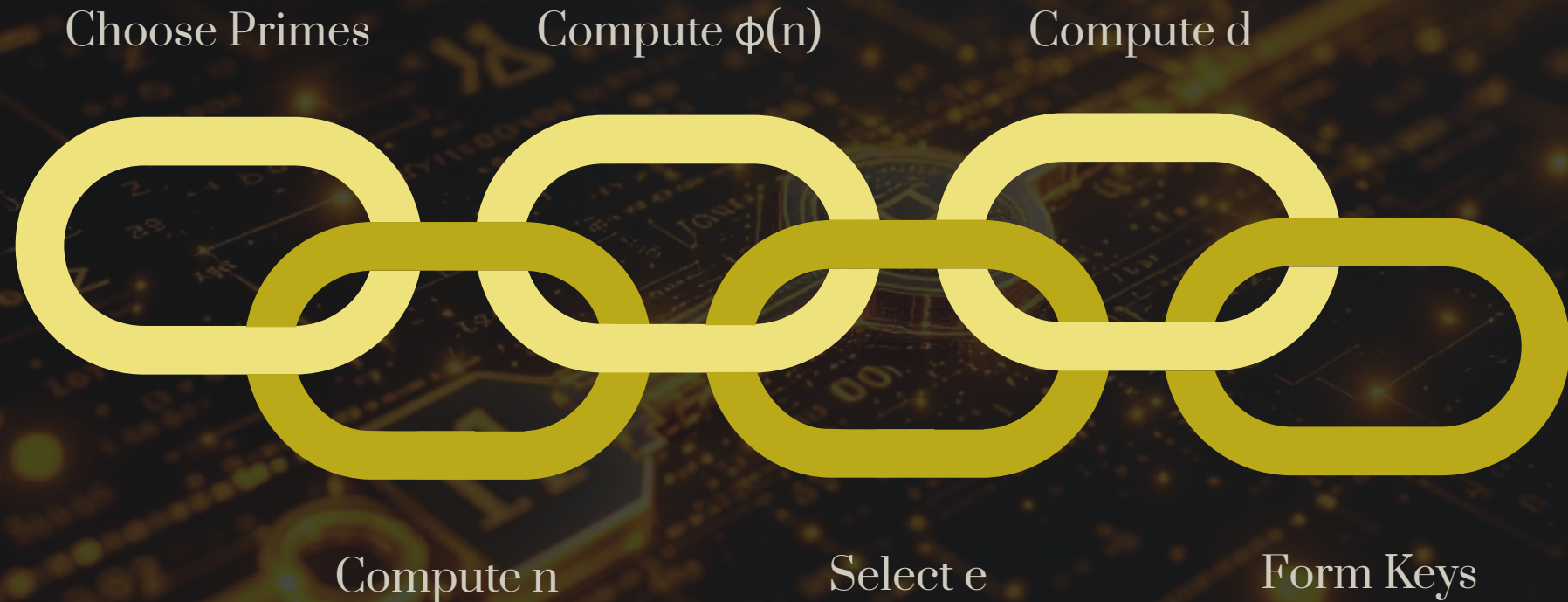
Euler's Theorem guarantees that repeated exponentiation will eventually cycle back to the original value, enabling the sender's encryption to be reliably reversed by the receiver.

From Theory to Practice



Creating the Keys

The strength of RSA encryption lies in the meticulous process of generating its public and private key pair. This sequence of steps ensures the mathematical properties required for secure communication.



This systematic approach forms the foundation for RSA's asymmetric encryption, where the public key can encrypt messages, but only the corresponding private key can decrypt them.

Step 1: Select Prime Numbers

The foundational step in RSA key generation is the careful selection of two very specific numbers:

Choose two large, distinct prime numbers, p and q .

For illustrative purposes, we will use smaller primes in our examples: $p = 61$, $q = 53$.

Why Large Primes Matter

The entire security of RSA encryption rests on the extreme difficulty of factoring the product of these two primes back into p and q .

In practical applications, these primes are massive, typically 2048 bits long, which corresponds to numbers with over 600 decimal digits.

Unbreakable by Design

Factoring such enormous numbers is computationally infeasible with current technology. It would take even the most powerful supercomputers millions of years to successfully factor a 2048-bit number.

Fun Fact: To ensure true randomness for their cryptographic keys, some companies like Cloudflare use a wall of [lava lamps](#) as a source of entropy!

Step 2 & 3: Calculate n and $\phi(n)$

With our chosen prime numbers, $p = 61$ and $q = 53$, we can now compute the two critical values that define our RSA system: the modulus n and Euler's Totient function $\phi(n)$.

Compute the Modulus n

The modulus n is the product of the two selected prime numbers. It will be used for all modular arithmetic in RSA.

$$n = p \times q = 61 \times 53 = \mathbf{3,233}$$

This n serves as the modulus for both the public and private keys in RSA. Its security relies on the computational difficulty of factoring it back into p and q .

Compute Euler's Totient $\phi(n)$

Euler's Totient function, $\phi(n)$, calculates the count of numbers less than n that are coprime to n .

$$\phi(n) = (p - 1)(q - 1) = (61 - 1)(53 - 1) = 60 \times 52 = \mathbf{3,120}$$

This value is crucial because it defines the order (size) of the multiplicative group Z_n^* within which all RSA operations occur. In our case, Z_{3233}^* has 3,120 elements.

Step 4 & 5: Public & Private Exponents

With the modulus n and Euler's Totient function $\phi(n)$ established, the next critical phase in RSA key generation involves carefully selecting the public exponent e and precisely computing its corresponding private exponent d .

Step 4: Choose Public Exponent e

- **Requirements:** $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
- **Our Choice:** For our example, we select $e = 17$.
- **Verification:** $\gcd(17, 3120) = 1$, confirming that 17 is coprime to 3120.

Common Choices: While $e = 3$ offers faster encryption, $e = 65,537$ is widely adopted in practice for its balance of security and computational efficiency.

Step 5: Compute Private Exponent d

Goal: Find d such that $ed \equiv 1 \pmod{\phi(n)}$.

Method: This crucial value is computed using the **Extended Euclidean Algorithm**, which efficiently solves for d .

For our example: We need to solve $17d \equiv 1 \pmod{3120}$.

Solution: The calculated private exponent is $d = 2,753$.

Verification: $17 \times 2,753 = 46,801 = 15 \times 3,120 + 1$. This confirms $17 \times 2,753 \equiv 1 \pmod{3120}$.

📌 **Security Note:** The private exponent d is kept **SECRET** – it is the cornerstone for secure decryption!

Step 6: Final Keys

With all calculations complete, we now assemble the Public and Private Keys, the essential components for secure RSA communication.



PUBLIC KEY

$$(e, n) = (17, 3233)$$

This key is meant to be shared openly. Anyone can use it to encrypt messages intended for you.



PRIVATE KEY

$$(d, n) = (2753, 3233)$$

This key must be kept absolutely secret. It is the only way to decrypt messages encrypted with your public key.

Metaphor: Your **Public Key** is like your mailbox address – anyone can send a letter to it. Your **Private Key** is the unique key to that mailbox – only you can open and read the letters.

Encryption & Decryption in Action

Let's walk through a complete example of RSA, encrypting a message with the public key and then decrypting it with the private key to recover the original.

Encrypting a Message

Using our message $m = 65$ and the public key $(e, n) = (17, 3233)$, we apply the encryption formula:

$$c \equiv m^e \pmod{n}$$

Substituting our values:

$$c \equiv 65^{17} \pmod{3233}$$

Through modular exponentiation (e.g., Square-and-Multiply), the calculated ciphertext is: **c = 2,790**

Decrypting the Ciphertext

Now, with the ciphertext $c = 2,790$ and the private key $(d, n) = (2753, 3233)$, we can recover the original message:

$$m \equiv c^d \pmod{n}$$

Substituting our values:

$$m \equiv 2790^{2753} \pmod{3233}$$

This complex calculation reveals the original message: **m = 65**



📌 **Success!** The RSA algorithm successfully transforms the original message into an unreadable form and then securely restores it, demonstrating its core function.

Why RSA Works: The Proof

The mathematical elegance of RSA comes to life in its decryption proof, which relies directly on Euler's Theorem to reliably recover the original message.

Mathematical Correctness

To recover the original message m from the ciphertext c using the private key (d, n) , we first define the relationship between e and d :

$$ed \equiv 1 + k\phi(n) \quad \text{for some integer } k \geq 0$$

When decrypting, we compute $m' \equiv c^d \pmod{n}$. Since $c \equiv m^e \pmod{n}$, we substitute to get:

$$m' \equiv (m^e)^d \pmod{n}$$

$$m' \equiv m^{ed} \pmod{n}$$

Substituting the expression for ed :

$$m' \equiv m^{1+k\phi(n)} \pmod{n}$$

$$m' \equiv m \cdot (m^{\phi(n)})^k \pmod{n}$$

By **Euler's Theorem**, if m is coprime to n , then $m^{\phi(n)} \equiv 1 \pmod{n}$. Applying this, we simplify:

$$m' \equiv m \cdot (1)^k \pmod{n}$$

$$m' \equiv m \pmod{n}$$

This demonstrates that the decrypted message m' is indeed congruent to the original message m .

📌 The Magic Behind the Math

The crucial step $m^{\phi(n)} \equiv 1 \pmod{n}$ (guaranteed by Euler's Theorem) is the reason RSA encryption and decryption are perfectly inverse operations, ensuring the original message can always be retrieved securely.



Part III: Why RSA is Secure

Group Theory Protects Your Data

The Hard Problems Behind RSA

The security of RSA encryption relies on the computational difficulty of solving specific mathematical problems. These "hard problems" are practically impossible to solve efficiently for sufficiently large numbers, making it infeasible for an attacker to break the encryption.



1. Integer Factorization

Given a large composite number n (the product of two prime numbers p and q), the problem is to find p and q . For RSA, n is extremely large, and no efficient algorithm exists to find its prime factors in a reasonable timeframe.



2. Computing $\phi(n)$ without p, q

To calculate Euler's Totient function $\phi(n)$, one typically needs the prime factors p and q of n . Without first factoring n , computing $\phi(n)$ directly is computationally equivalent to factoring n itself.



3. Finding d without $\phi(n)$

The private exponent d depends on $\phi(n)$, as it is the modular multiplicative inverse of the public exponent e modulo $\phi(n)$. Therefore, if $\phi(n)$ cannot be efficiently computed, then d cannot be efficiently derived either.

How Group Structure Protects RSA

The security of RSA is fundamentally rooted in the properties of group theory, specifically the characteristics of the multiplicative group of integers modulo n , Z_n^* .

Hidden Group Order

The size of the group, $\phi(n)$, is known only to the key owner, preventing attackers from deriving the private exponent d .

One-Way Exponentiation

Encrypting messages is easy, but decrypting without the private key is computationally extremely difficult, making it a one-way function.

Inverse Requires Secrecy

The private exponent d , crucial for decryption, can only be calculated if the attacker can factor the large number n .

❏ Bottom Line:

Group theory isn't just math – it's the invisible shield that protects your digital transactions and communications.

Bigger Keys = Better Security

The strength of RSA encryption is directly tied to the size of the prime numbers chosen, which determines the length of the public and private keys. Larger keys make it exponentially more difficult for attackers to factor n and break the encryption.

Here's a breakdown of common RSA key sizes and their security implications:

512 bits	Broken	Hours
1024 bits	Deprecated	Months
2048 bits	Current Standard	Millions of years
3072 bits	High Security	Unfeasible
4096 bits	Maximum Security	Impossible

Recommendation:

For robust security, always use **at least 2048-bit RSA keys**. Many organizations are now moving towards 3072-bit or higher for long-term protection.

There's a trade-off: **larger keys provide greater security but require more computational power, leading to slower encryption and decryption processes**. Organizations must balance security needs with performance considerations.

Part IV: RSA in the Real World

Protecting Your Digital Life

RSA Secures Your Money

RSA is the cornerstone of trust in online banking, ensuring that your financial data and transactions remain confidential and protected from unauthorized access.



Login Authentication

RSA encrypts your passwords during transmission and secures One-Time Password (OTP) protection, safeguarding your access to accounts.



Transaction Security

RSA facilitates a hybrid encryption approach, securely exchanging session keys that AES then uses to encrypt sensitive transaction data.



ATM Communications

RSA certificates authenticate ATMs, preventing counterfeit machines and securing PIN transmission, complying with standards like PCI DSS.

Shop Safely with RSA

RSA is integral to securing e-commerce, safeguarding sensitive financial data and verifying the authenticity of online merchants.



Payment Processing

RSA encrypts your credit card details and secures communication with payment gateways, preventing interception during transactions.



SSL/TLS Certificates

The green padlock in your browser signifies an RSA-secured connection, verifying the website's identity and encrypting all data exchanged.



Digital Signatures

RSA-based digital signatures confirm merchant legitimacy and prevent transaction tampering, ensuring your approvals are processed accurately.

📄 **Over 2 billion websites rely on RSA-based TLS to protect their connections**, making it a foundational element of internet security.

Modern Alternatives: RSA vs. ECC

While RSA remains a cryptographic cornerstone, other modern alternatives like Elliptic Curve Cryptography (ECC) offer compelling advantages, particularly in resource-constrained environments.

Feature	RSA	ECC
Key Size	2048-4096 bits	256-521 bits
Speed	Slower	Faster
Security/bit	Lower	Higher
Best For	Legacy systems, certificates	Mobile, IoT, modern apps
Complexity	Simple math	Complex curves
Adoption	Universal	Growing rapidly

Recommendation:

- Use **RSA-2048 for compatibility** with existing infrastructure.
- Leverage **ECC for new mobile/IoT systems** and applications requiring high security with smaller key sizes and faster performance.
- Consider **hybrid approaches** for optimal flexibility and security.

RSA by the Numbers

The impact of RSA in securing our digital world is immense, reflecting its foundational role across various domains.

2B+

Websites Protected

Utilizing RSA-based encryption for secure communication.

\$10T+

Annual Transactions

Safeguarded through RSA in online banking and e-commerce.

99.9%

Secure Connections

Rely on RSA or its derivatives for establishing trust online.

617

Digits per Prime

The approximate length of prime numbers in 2048-bit RSA keys.

Millions

Years to Break

Estimated time to factor large RSA keys with current computers.

40+

Years of Trust

RSA has maintained cryptographic integrity since its inception (1977).

RSA in Action: A Live Demonstration

Let's bring the theory to life with a simple Python script that illustrates the core steps of RSA: generating keys, encrypting a message, and decrypting it back to the original form. This interactive example uses small numbers for clarity, but the underlying principles scale to the massive primes that secure real-world data.

Further Reading

Deepen your understanding of RSA and its mathematical underpinnings with these essential resources.

Key Papers

- Rivest, Shamir & Adleman (1978)
- Original RSA paper
- Katz & Lindell - Introduction to Modern Cryptography
- NIST PQC Standardization Project

Online Resources

- Cloudflare Blog on cryptography
- Khan Academy: Modular arithmetic
- Computerphile: RSA video explanations

Textbooks

- Dummit & Foote: Abstract Algebra
- Stinson: Cryptography Theory and Practice

Questions?

Please feel free to ask anything you'd like to clarify about Group Theory, RSA, or our presentation.

Our Team

- Geetika Pachauri
- Anusha Anindita
- Harshil Pansala
- Shirsha Pattanaik
- Pearl Mendapara

Faculty Advisor

Professor Pradeep Kumar, Department of Mathematics, Mahindra University

Thank you for your attention!

RSA in Our Digital Lives

RSA's influence extends across virtually every facet of our digital world, underpinning the security and trust essential for modern online interactions.



Banking

Secure financial transactions and protected ATM communications.



E-commerce

Ensures payment protection and encrypted data exchange via SSL/TLS.



Email

Provides digital signatures for authenticity and secure email authentication.



Web

Secures website connections (HTTPS) and validates digital certificates.



Mobile

Protects mobile applications and facilitates secure messaging.

RSA is used in 95% of secure internet connections worldwide.